Impact of stale calibration on tail risk estimates

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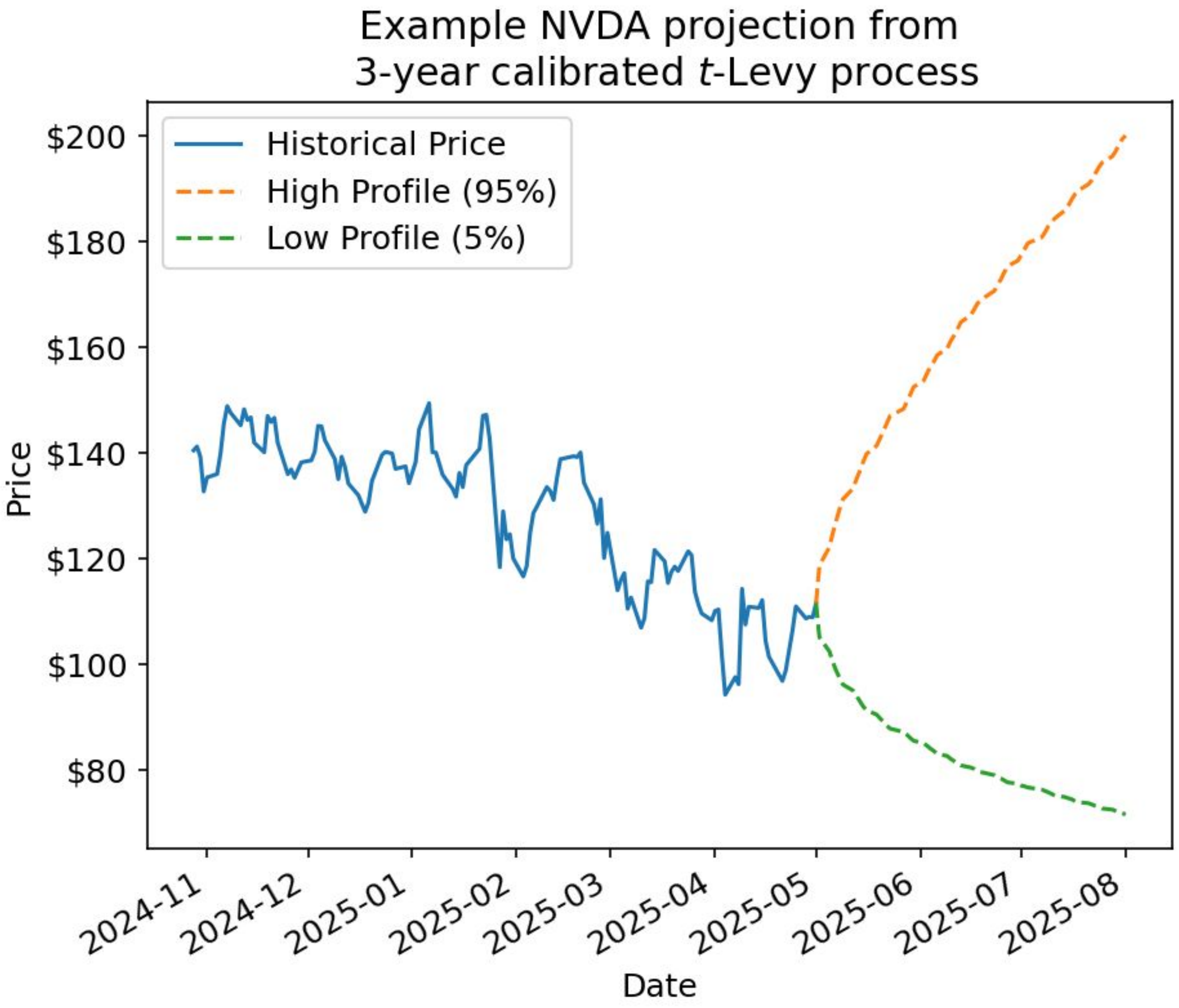
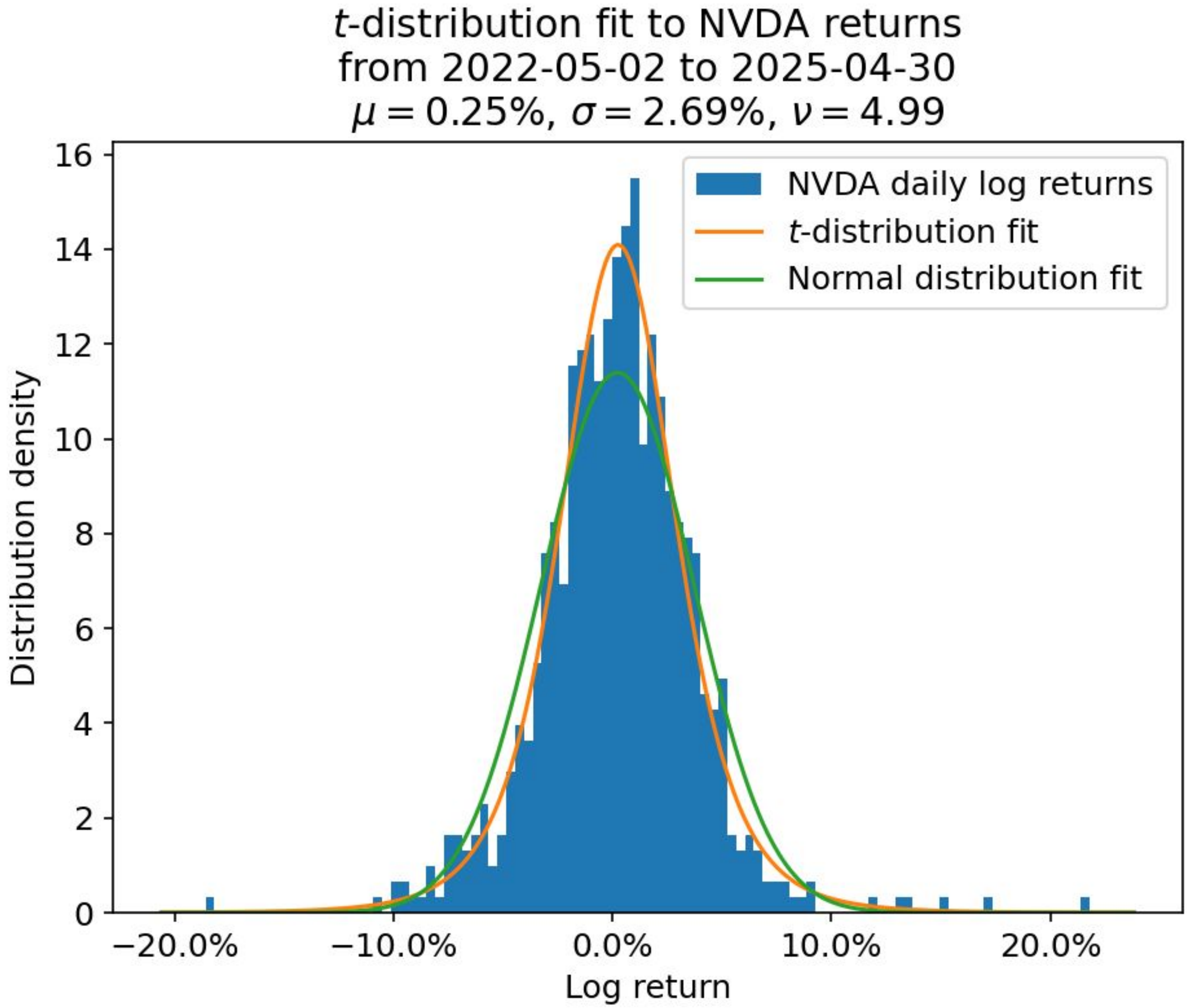
Background

- Quants frequently create risk models that forecast confidence intervals of future asset prices
- **Ideally**, such models could be recalibrated frequently
- **In practice**, recalibrations often have to adhere to technology release schedules or may be delayed due to unforeseen issues
- We want to explore the impact of such a "stale" calibration



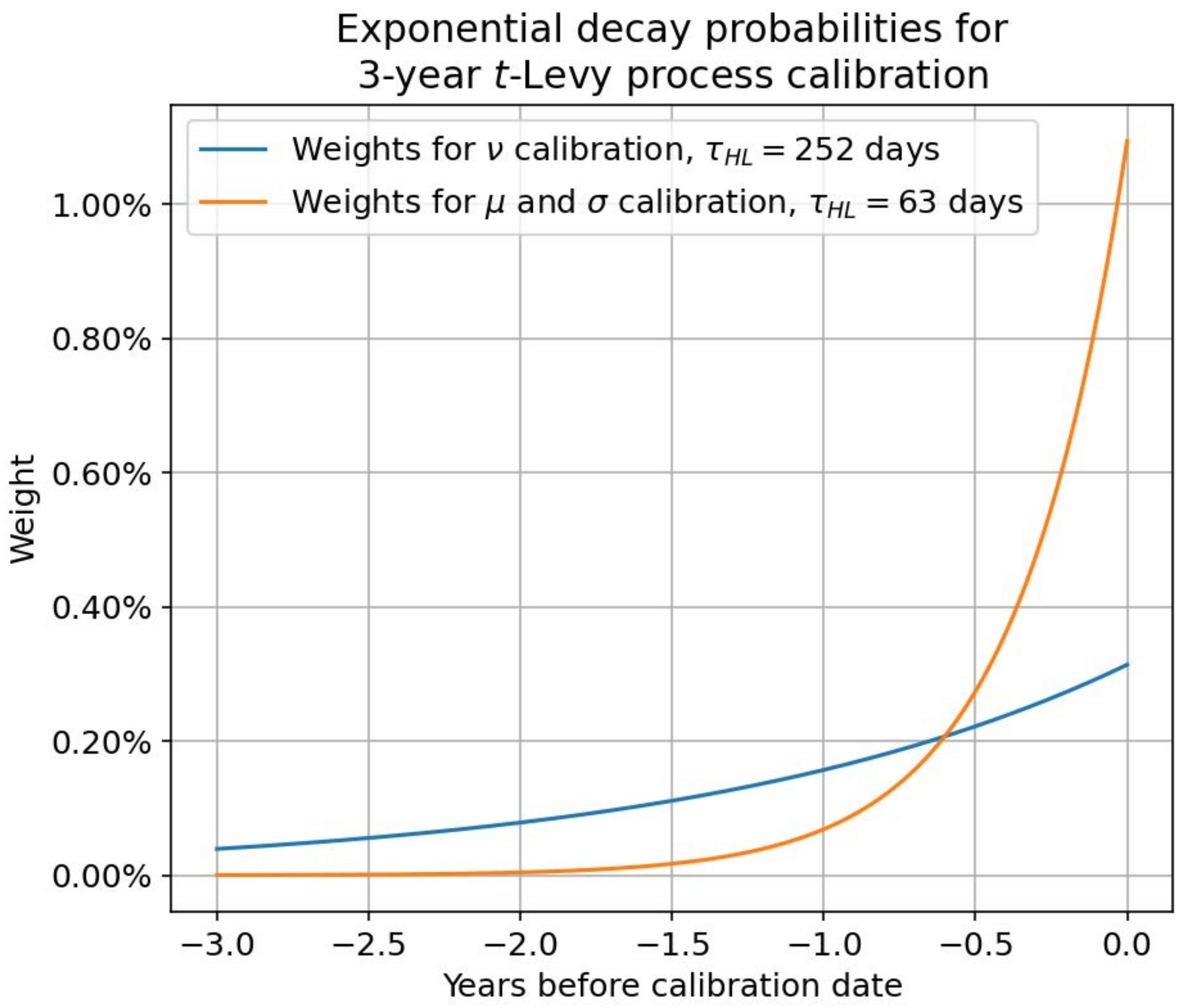
t-distribution Levy process calibration

- My focus is on 2-week price changes and 3-year calibrations
- Model daily changes of stock log-price as i.i.d. samples of a t-distribution
    - $\Delta \boldsymbol{X}_t \sim t(\boldsymbol{\mu}, \boldsymbol{\sigma}^2, \nu)$
- Determine t-distribution parameters by a maximum likelihood fit to historical data
- We'd prefer to weight recent data more heavily than older data

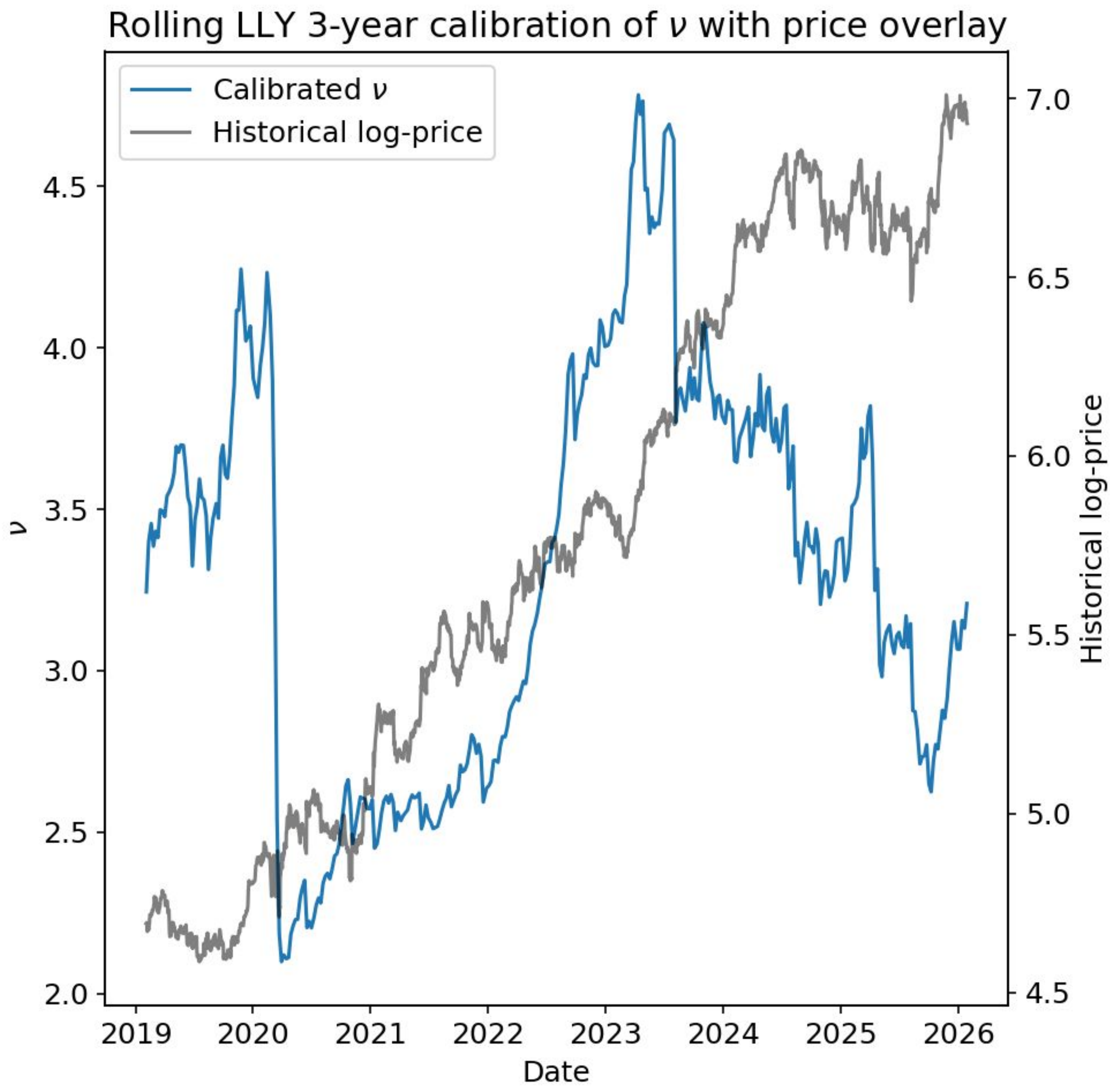


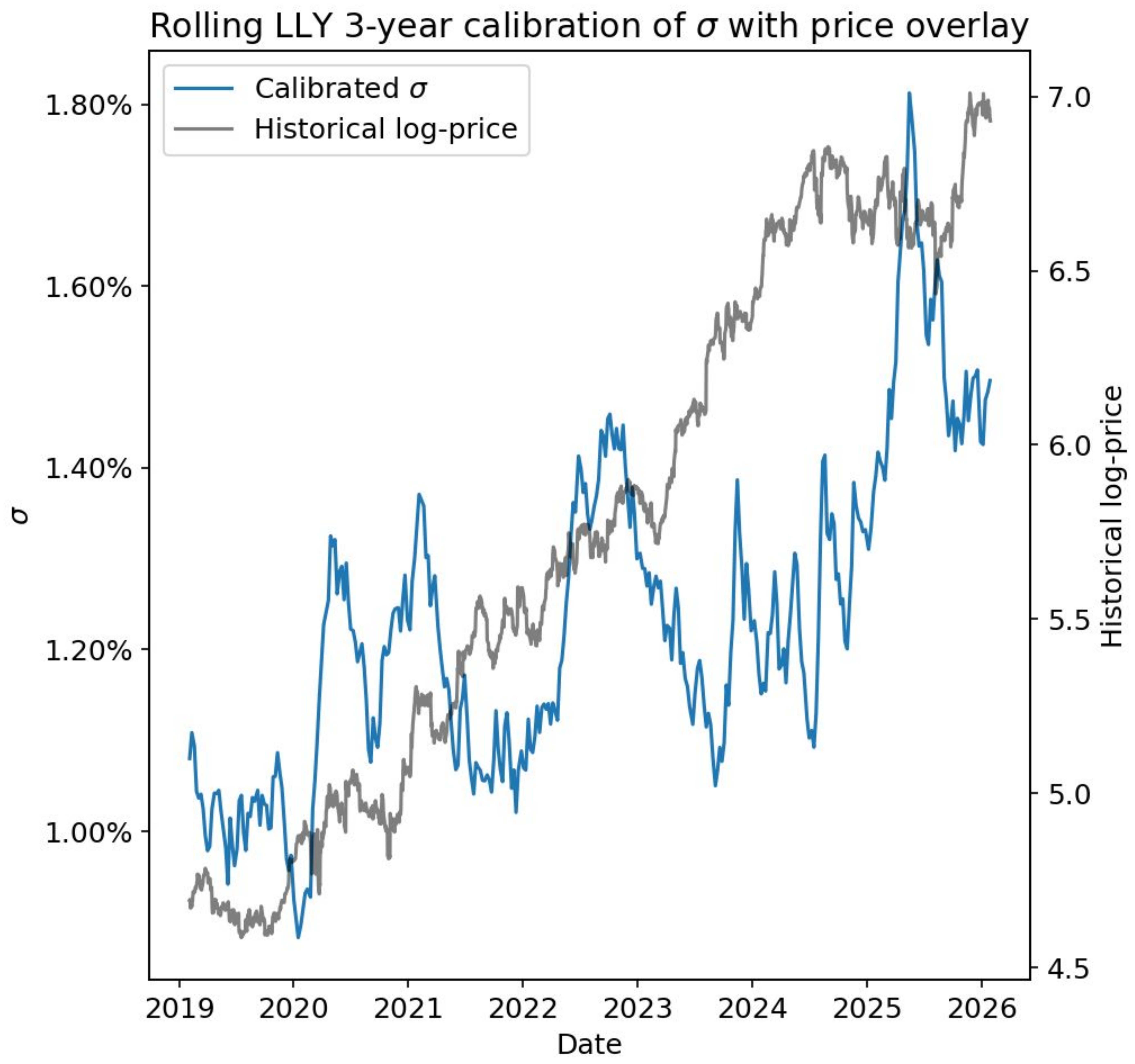
Exponential decay flexible probabilities

- Weight historical data so that weight decays by half each half-life
  - $$p_t|\tau_{\mathrm{HL}} \equiv \frac{e^{-\frac{\ln(2)}{\tau_{\mathrm{HL}}}|\bar{t}-t|}}{\sum_s e^{-\frac{\ln(2)}{\tau_{\mathrm{HL}}}|\bar{t}-s|}}$$
- Choice of half-life is a **trade-off** between calibration responsiveness and model stability
- Degrees of freedom estimation is more inherently unstable and requires longer half-lives



Stability of parameter estimates





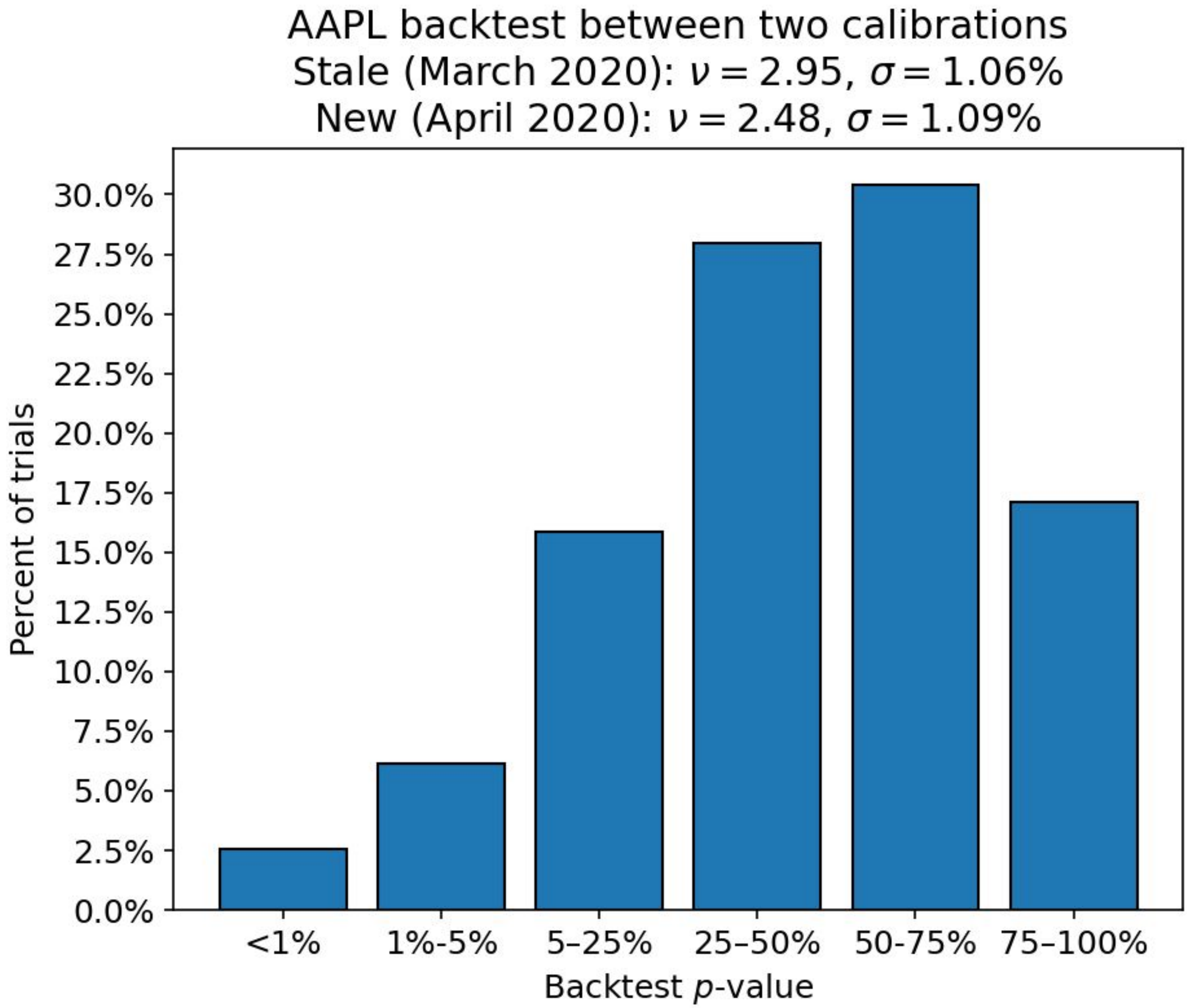
Basic backtesting

- Given our model predicts a confidence interval on future asset prices, we can use a simple binomial test to measure performance
  - Historical exceedance of our predicted confidence level is a “failure”
- With ~25 non-overlapping 2-week samples per year, this is treated more as a monitored metric than a proper statistical test

Failures <= k	Binomial CDF with 25 samples and 95% probability of success
0	100.0%
1	72.3%
2	35.8%
3	12.7%
4	3.4%
5	0.7%

Discussing stale calibration through backtesting impact

- A simple method of assessing stale calibrations is through the **impact to the backtesting measure**
  - Treat newly calibrated model as a ground truth to compare stale calibration to
- Using a stale calibration in place of an updated one **effectively shifts the failure threshold** of the backtest



Rough extrapolations of parameter drift over time

- One could also extrapolate out historical trends in the calibration parameters
- E.g., on average, for AAPL in the last 7 years,
  - “Decreasing degrees of freedom tend to change -0.22 per month”
  - “Increasing mu / sigma tend to change +0.064% / +0.088% per month”
  - **Conservative approximation assuming a *lot* of independence!**
- Parameters are stable in the short term, and meaningful backtesting impact appears over months